

Derek Wang

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EDUCATION

University of Waterloo

Expected Apr 2030

Bachelor of Applied Science in Computer Engineering (Co-op)

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Prandtl Dynamics Inc.

Richmond Hill, ON

- Enabled real-time tracking of 5 simultaneous moving targets at 170m range and 20ms latency by integrating SRP-PHAT acoustic beamforming with UKF + Global Nearest Neighbor multi-sensor fusion
- Cut false alarms by 40% and halved model size at 98% mAP by migrating audio detection to EfficientAT
- Eliminated cross-tenant data exposure by org-scoping queries across 6 dashboard services and WebSockets
- Hardened auth and supply chain with fail-closed guards, timing-safe key checks, and CI audit gating
- Slashed recording setup from 10 to 2 minutes by replacing a tethered rig with a Raspberry Pi service that buffers audio in memory and syncs WAV + JSON metadata into SQLite over SSH for downstream model training

Software Engineer Intern

March 2026 – June 2026

Kenoxis Technologies Inc.

Mississauga, ON

- Cut API latency by 35% with Redis caching across Node.js microservices on a fraud detection platform
- Designed MongoDB schemas and ingestion pipelines for job postings, improving refresh from daily to hourly
- Integrated LLM inference into backend services via async queues, reducing time-to-flag suspicious postings by 60%
- Built candidate distance and map APIs using MongoDB geospatial queries to filter applicants by location
- Accelerated onboarding of 5+ new interns by authoring internal API documentation and integration guides

Autonomous Self Driving Software Engineer

Sept 2025 – Present

WATonomous – Student Design Team

Waterloo, ON

- Architected a ROS2 navigation stack running A* path planning over costmap grids and Pure Pursuit control under 10ms per cycle, enabling real-time autonomous rover navigation across dynamic terrain (hills, obstacles, holes)
- Led YOLO11s model retraining, improving detection mean average precision from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Optimized YOLO11s inference to 24 FPS on rover hardware via ONNX export and onnxslim optimization

PROJECTS

Reparo - AI Repair Tool (Hack Canada 2026 Winner) | *Python, FastAPI, Swift, Gemini, SerpAPI, Reactiv*

- Generated 5+ structured repair outputs from a single device image using a computer-vision + Gemini reasoning pipeline that performs component detection and contextual repair inference for consumer devices
- Built a FastAPI orchestration service applying object-oriented design to normalize unstructured LLM responses into typed repair models, enabling deterministic UI rendering of repair steps, cost estimates, and required tools
- Designed a multi-stage parts retrieval pipeline (SerpAPI → Shopify storefront discovery → Shopify product APIs) to automatically source purchasable repair components surfaced in a Swift Reactiv App Clip

Silhouette - Cursor for Fashion | *Node.js, Next.js, Express, MongoDB, Gemini, Overshoot.ai SDK*

- Delivered personalized fashion recommendations in under 2s by building Gemini pipelines with MERN stack that orchestrate async inference, product recommendations, and style constraints over Next.js live camera input
- Secured a VC referral to continue development as a Proof of Concept (Tailor AI) for Merian Brothers

Github Repository Analyzer | *Node.js, Express, React, Gemini, Octokit, Puppeteer*

- Designed and shipped a full-stack AI-powered GitHub repository analyzer that reduced codebase evaluation time by 98% to under 90 seconds per repository, making technical repos accessible to non-technical stakeholders
- Integrated Octokit and Puppeteer to automatically extract repo metadata, file trees, and source code for analysis
- Developed a 3-agent Gemini AI pipeline generating 3 different audience-specific reports from a single codebase scan

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Frameworks: React, Next.js, Node.js, Express, FastAPI, Vite

Cloud/DevOps: AWS, Azure, Docker, GitHub Actions

Libraries/Tools: PyTorch, OpenCV, SQLite, PostgreSQL, MongoDB, Redis, Git, ROS2, Foxglove, Fusion 360